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The Polarization Paradox: Suppression Filters, Exit, and Active-User Polarization

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Abstract: Ideological polarization on social media can harm both discourse and platform retention through user exit. Platforms often respond with suppression-oriented feed filters that reduce exposure to confrontational content in Social Judgment Theory's latitude of rejection. We study long-run effects with Dynamic-BABE, a stylized ABM combining biased assimilation, social-judgment zones, exit, and co-evolving dyadic trust. A 2×2 design crosses the Suppression Filter with trust co-evolution (30 replications per cell; $N = 500$, 200 steps). The Polarization Paradox is a retention-versus-survivor-dispersion trade-off—a KPI misalignment, not population warming: the filter cuts cumulative churn from 39.18% to 0.38% while active-agent polarization rises from 0.9550 to 0.9904 because extremes who would otherwise exit remain communicating; full-cohort σ_{all} is unchanged (0.9904 vs. 0.9905). Step-wise revenue tracks that retention under a pre-polarized cliff that never switches. Enabling trust alone produces trust segregation (mean ratio 31.28; median 26.76), a separate axis that does not define the filter paradox. Seed-paired Filter × Trust tests show no evidence that trust resolves the active-set trade-off. In short, suppressing disagreement can protect retention without lowering active-set dispersion among survivors.

Keywords: agent-based modeling; opinion dynamics; polarization; dyadic trust; algorithmic curation; social media

Introduction

1.1 Digital platforms now structure much of the public sphere. They also concentrate disagreement: algorithmic feeds can amplify conflict and deepen affective polarization (Bail et al. 2018; Levy 2021).

1.2 For platforms, polarization is an economic problem as well as a social one. Hostile feeds raise cognitive load; users may leave. Hirschman (1970) called this exit. Exit shrinks the active audience and advertising impressions. We report step-wise advertising revenue as a readability transform of retention under a stylized constant unsafe multiplier: in the main design the brand-safety cliff never switches, so dollar contrasts track active users rather than a changing safety regime (Guess et al. 2023; Levy 2021).

1.3 Platforms therefore experiment with structural feed interventions that reduce exposure to confrontational content before conflict escalates. We model one such intervention as a probabilistic Suppression Filter (short: the filter). It intercepts interactions in the latitude of rejection (Sherif & Hovland 1961)—messages so far from the receiver's position that exposure tends to trigger modelled backlash rather than assimilation. This is not constructive bridging ranking that promotes cross-camp contact; it suppresses high-friction exposure. Field experiments (Guess et al. 2023; Bail et al. 2018) show that exposure regimes shape attitudes and behavior; our filter is an abstraction of suppression-oriented curation, not a claim about any specific product.

1.4 Suppression can create a trade-off we call the Polarization Paradox: the filter sharply improves retention (and therefore step-wise revenue under our stylized scale) while active-agent polarization stays high or rises because extreme agents who would otherwise exit remain in the communicating population. Full-cohort polarization σ_{all} (churned agents retain last opinions) does not show that rise, so the paradox is a retention-versus-survivor-dispersion / KPI-misalignment result rather than a warming of the entire population. In our main 2×2, initial polarization already sits above the stylized brand-safety cliff ($\sigma(0) \approx 0.7 > 0.6$), so the revenue contrast is demographic.

1.5 We formalize this with Dynamic-BABE (Biased Assimilation & Behavioral Entrenchment). Closest formal ancestors include Chen-style biased assimilation (Chen et al. 2021), bounded-confidence opinion dynamics (Hegselmann & Krause 2002; Deffuant et al. 2000), Social Judgment repulsion (Sherif & Hovland 1961; Jager & Amblard 2005), and algorithmic-bias / selection models that amplify fragmentation without an exit channel (Sîrbu et al. 2019). Relative to those cited baselines, to adaptive-network rewiring (Macy & Willer 2002; Keijzer et al. 2018), and to feed-ranking field experiments (Guess et al. 2023; Levy 2021), we contribute a coupled suppression-filter×exit design in which retention and active-set dispersion can diverge—with σ_{all} as a required honesty check—plus a separate co-evolving dyadic trust axis on a fixed topology. That is a contrast to the neighbours we cite, not a claim of field-wide uniqueness. Following Slovic (1993), trust builds slowly under agreement and erodes quickly under conflict.

1.6 A 2×2 experiment crosses the Suppression Filter with dynamic trust. The filter paradox is identified with Trust OFF (Baseline vs. Filter Only). Separately, trust co-evolution produces trust segregation: trust concentrates within opinion camps while outgroup trust collapses. That pattern is distinct from the filter paradox; under our defaults we find no evidence that trust moderates the filter×exit trade-off (non-significance is not equivalence).

1.7 Our contributions are threefold. First, Dynamic-BABE links cognition, dyadic trust, exit, and a stylized retention readout in one ABM. Second, we document the Polarization Paradox as a filter×exit survivorship / KPI-misalignment effect on active-user polarization (with σ_{all} as a required robustness check). Third, we argue that retention metrics alone are incomplete if the goal also includes active-set dispersion and local same-pole structure among remaining users.

1.8 Replication materials are available at `https://github.com/TheCoolSam/BABEModel`; we intend to deposit the model on CoMSES Net for permanent archival.[1]

Related Work

2.1 We situate Dynamic-BABE in five literatures: classical opinion dynamics, biased assimilation, Social Judgment Theory, co-evolving trust, and algorithmic exposure.

Classical Models of Opinion Dynamics

2.2 DeGroot (1974) modelled influence as repeated weighted averaging. In a connected network this process converges to consensus, which cannot explain persistent disagreement. Bounded-confidence models restrict influence to neighbours within a similarity threshold (Hegselmann & Krause 2002; Deffuant et al. 2000) and can produce multiple clusters. Flache et al. (2017) review these and related formalisms. Bounded confidence typically ignores out-of-range neighbours rather than modelling active repulsion, and classical OD usually lacks an exit channel that changes the communicating population.

Cognitive Resistance and Biased Assimilation

2.3 Lord et al. (1979) showed that people can interpret mixed evidence in ways that strengthen prior attitudes—biased assimilation. Dandekar et al. (2013) embedded related biases in network models and linked them to polarization under homophily. Chen et al. (2021) proposed BEBA, where influence weights $w_{ij}(t) = 1 + \beta_i y_i(t) y_j(t)$ depend on entrenchment β_i and alignment, so negative influence can emerge endogenously. Mäs & Flache (2013) discuss bipolarization without explicit distancing. We adapt Chen-style weights to pairwise, multi-issue interactions, map them onto Social Judgment zones (Sherif & Hovland 1961; Jager & Amblard 2005), and add co-evolving dyadic trust.

Social Judgment Theory and Backfire

2.4 Social Judgment Theory partitions attitude space into latitudes of acceptance, non-commitment, and rejection (Sherif & Hovland 1961). Messages in the acceptance latitude tend to pull opinions toward the source; messages in the rejection latitude can produce contrast and backlash. Empirical evidence for attitude backfire / boomerang effects is contested: some exposure regimes raise extremity (Bail et al. 2018), while feed and correction experiments often find null or highly context-dependent attitude change (Guess et al. 2023; Levy 2021). We treat Zone 3 repulsion as a modelled Social Judgment mechanism, not as an empirical universal. Jager & Amblard (2005) showed that repulsion in networks can create opposing camps. We treat influence weights as determining these zones.

Networks and Dyadic Trust

2.5 Opinions and social structure often co-evolve (Macy & Willer 2002; Centola 2010; Keijzer et al. 2018). Many models allow rewiring toward similar others. We hold the undirected graph fixed and instead let edge trust change with interaction. Trust is asymmetric in gain and loss (Slovic 1993): it rises slowly after agreement and falls faster after conflict. High trust buffers occasional disagreement; low trust makes further conflict more likely.

Algorithmic Interventions, Filtering, and Exposure

2.6 Platforms once hoped that cross-cutting exposure would reduce polarization. Bail et al. (2018) found that exposing Twitter users to opposing-party bots could increase extremity. Feed experiments also show that ranking regimes change on-platform behavior, often more than deep attitudes (Levy 2021; Guess et al. 2023). On the modelling side, algorithmic bias in partner selection can amplify fragmentation and polarization in bounded-confidence settings without modelling exit (Sîrbu et al. 2019); adaptive-network work often rewires ties (Macy & Willer 2002; Keijzer et al. 2018). Relative to those cited neighbours, Dynamic-BABE couples a suppression filter to Hirschmanian exit so that retention can rise while active-set σ stays high or rises, with σ_{all} showing that the rise is survivorship rather than population warming, and adds a separate trust-segregation axis on fixed topology. We do not claim that no prior model has touched filtering, exit, or trade-offs in isolation—only that this joint packaging differs from the baselines we engage. We abstract suppression as a probabilistic filter that intercepts rejection-zone events—distinct from ranking that promotes like-minded contact (Sîrbu et al. 2019) or constructive bridging. We use "Polarization Paradox" for that retention-versus-survivor-dispersion / KPI-misalignment tension without claiming uniqueness of the name. Field studies rarely observe long-run feedback among opinions, trust, and exit; the ABM is built for that gap.

Model Description (ODD+D Protocol)

3.1 We describe Dynamic-BABE with the ODD+D protocol (Grimm et al. 2010; Müller et al. 2013) so that entities, processes, and decision rules are explicit and reproducible.

Overview

Purpose

3.2 Dynamic-BABE is a theoretical laboratory for feedback among interpersonal trust, biased assimilation / rejection-zone repulsion, and a stylized Suppression Filter. We use it to study the Polarization Paradox: the filter can cut exit and protect retention-scaled revenue while leaving (or raising) active-agent polarization among survivors; full-cohort σ_{all} need not rise.

Entities, State Variables, and Scales

3.3 Two entities drive the simulation: agents (users navigating their feeds) and the network environment (the network's structural pathways and the platform's moderation layer).

Agents:

3.4 Each agent $i \in \{1, 2, \ldots, N_{\text{agents}}\}$ represents a single user, tracked via several state variables:

- Opinion Vector ($\vec{O}_i \in [-1, 1]^K$): A multi-dimensional ideological signature representing the agent's stance on K distinct social or political issues. The issue dimension $d \in \{1, \ldots, K\}$ spans continuously from -1.0 (progressive/left) to $+1.0$ (conservative/right).
- Salience Vector ($\vec{S}_i \in [0, 1]^K$): An attention allocation vector mapping the relative cognitive importance the agent assigns to each issue. We constrain this vector so that salience sums to unity: $\sum_{d=1}^{K} S_{i,d} = 1.0$.
- Cognitive Entrenchment ($\beta_i \in [\beta_{\min}, \infty)$): A scalar tracking cognitive resistance, dogmatism, or confirmation bias (Chen et al. 2021). Elevated β_i values amplify outgroup rejection and fuel stronger backfire repulsion.
- Behavioral Frustration ($F_i \in \mathbb{N}_0$): An integer counter tracking the psychological irritation and cognitive dissonance that mount during confrontational, cross-cutting exchanges.
- Active Status ($\text{Active}_i \in \{\text{True}, \text{False}\}$): A boolean flag marking platform activity. When frustration strictly exceeds T_c ($F_i > T_c$), the agent permanently exits ($\text{Active}_i = \text{False}$); with default $T_c = 15$, exit occurs at $F_i = 16$.

Network and Platform Environment:

3.5 We model the social topology as an undirected, scale-free graph $G = (V, E)$ generated via the Barabási-Albert (BA) preferential attachment algorithm (Barabási & Albert 1999). The node set V represents agents, while the edge set E maps communication pathways.

- Dyadic Interpersonal Trust ($T_{ij} \in [0, 1]$): A symmetric weight on edge (i, j) when Trust ON, initialized at $T_0 = 0.5$. Trust OFF omits the trust buffer from w_{ij} entirely (edges are not trust-initialized).
- Algorithmic Interception Layer: When Filter ON, the platform intercepts a fraction of Zone 3 events and applies frustration relief.

Scales:

3.6 Time ticks in discrete steps $t \in \{1, 2, \ldots, t_{\max}\}$, where each step represents a conversational epoch. Space is purely topological, dictated by network connections. The model populates a scale-free network with $N_{\text{agents}} = 500$ agents and preferential attachment parameter $m_e = 3$, yielding degree distributions that mirror real-world social networks.

Process Overview and Scheduling

3.7 Each step t triggers a sequence of operations that update opinions, frustration, trust, and churn:

1. Randomized Activation Order: Mesa 2.x `RandomActivation` shuffles agents using `model.random`, seeded via `reset_randomizer(seed)`. Partner choice and filter draws use a separate `numpy.Generator(seed)`.
2. Agent Interaction Loop: We iterate through each active agent i in the shuffled list:
   (a) The agent filters its neighbors to isolate active peers: $\mathcal{N}_i^{\text{active}}(t) = \{j \in \mathcal{N}_i : \text{Active}_j(t) = \text{True}\}$. If this neighborhood is empty, the agent skips its turn.
   (b) The agent selects a dialogue partner j from $\mathcal{N}_i^{\text{active}}(t)$ uniformly at random ($k_{\text{step}} = 1$).
   (c) The directed pipeline $\text{Interact}(i, j)$ updates only receiver i's opinion and frustration; symmetric trust T_{ij} updates when Trust ON.
3. Passive Trust Decay: After all interactions, decay applies to every edge, including edges updated in the same step (so a just-updated tie also decays within the tick):

$$T_{uv}(t+1) = T_{uv}(t) \cdot (1 - \lambda) \tag{1}$$

4. Platform Diagnostics: The model computes and logs global metrics: opinion polarization $\sigma(t)$ over active agents, full-cohort polarization $\sigma_{\text{all}}(t)$ (churned agents retain last opinions), cumulative churn, user frustration, active user count, ingroup/outgroup trust divergence, and same-pole triangle purity.

5. Termination Check: The simulation halts when the step count reaches $t_{\max} = 200$, or if all users exit ($N_{\text{active}} = 0$).

Design Concepts

Theoretical Background

3.8 We anchor Dynamic-BABE's cognitive architecture in Social Judgment Theory (Sherif & Hovland 1961; Jager & Amblard 2005). This framework partitions an agent's attitude spectrum into three distinct zones: the Latitude of Acceptance (yielding assimilative agreement), the Latitude of Non-Commitment (resulting in inertia), and the Latitude of Rejection (triggering backfire repulsion).

3.9 To formalize cognitive entrenchment (β_i) and opinion alignment (A_{ij}), we adapt the Biased Assimilation and Behavioral Entrenchment (BEBA) model proposed by Chen et al. (2021), which extends DeGroot's classical consensus dynamics (DeGroot 1974). We merge this with a dynamic trust layer inspired by Slovic's empirical research (Slovic 1993). This integration captures the fundamental asymmetry of trust: relational capital accumulates painfully slowly through agreement but collapses instantly during conflict.

3.10 Finally, we operationalize user exit by mapping the accumulation of frustration to the classic Exit-Voice-Loyalty (EVL) framework of Hirschman (1970).

Individual Decision-Making

3.11 Agents employ heuristic cognitive updates rather than rational utility maximization, strictly following the bounded rationality principles of the ODD+D protocol.

- Objectives: Agents do not maximize an explicit utility or payoff function. Partner choice is uniform at random among active neighbors; there is no selective seeking of agreeable contacts. After contact, zone-conditioned updates (assimilation, ignore, or repulsion) and frustration dynamics implement a heuristic response that reduces dissonance when agreement occurs and raises exit pressure under repeated rejection-zone conflict.

- Information Processing: When active, an agent perceives the opinion vector $\vec{O}_j$ and salience weights $\vec{S}_j$ of their chosen dialogue partner. They process this information heuristically by calculating their salience-weighted opinion alignment A_{ij} and checking whether the resulting influence weight lands in their Latitude of Acceptance, Non-Commitment, or Rejection.

- Adaptive Behavior: Agents adapt dynamically across two scales. Cognitively, they adjust their opinion vectors $\vec{O}_i$ to assimilate toward agreeable peers or repel away from hostile ones. Behaviorally, they accumulate frustration F_i when clashing with others, leading to permanent exit (churn) if their frustration threshold is breached. Relationally, their dyadic trust T_{ij} adapts to reward agreement and penalize conflict, reshaping future interaction weights.

Learning

3.12 Although agents lack formal reinforcement learning capabilities, they maintain a relational memory via dyadic trust (T_{ij}). They continuously update their assessment of neighbor compatibility. Agreeable, assimilative dialogue slowly accumulates trust, whereas hostile backfires erode it quickly. This co-evolving relational memory shapes future influence weights, closing the feedback loop between network relations and cognitive updates.

Individual Sensing

3.13 Each agent i tracks its direct topological neighbors with perfect accuracy. During dialogue, agent i observes a neighbor's opinion vector $\vec{O}_j$ and salience weights $\vec{S}_j$. Agents remain blind to the state variables, cognitive profiles, and degrees of non-neighboring nodes, and they hold zero global knowledge regarding system-level polarization or platform-level financials.

Individual Prediction

3.14 Agents react purely to immediate events, operating without forward-looking predictions. They neither anticipate neighbor opinion shifts nor forecast their own impending exit as frustration mounts toward the churn threshold.

Interaction

3.15 Interactions unfold locally, dyadically, and directionally. While trust updates symmetrically across the edge (i, j), opinion and frustration updates execute from the perspective of the active agent receiving the communication. The platform's suppression filter acts as an external gatekeeper, intercepting high-friction exchanges to suppress the behavioral symptoms of disagreement.

Collectives

3.16 No formal collectives (such as political parties or coalitions) exist in our model. However, functional collectives emerge endogenously through conversational dynamics. Agents coalesce into two opposing ideological factions based on their opinion signs. These emergent camps govern the flow of trust and drive the self-organization of network clusters.

Emergence

3.17 Macro patterns that are not hard-coded include: (i) the Polarization Paradox—high retention with high active-set σ when the filter blocks exit of extremes, while σ_{all} stays flat; (ii) trust segregation when dyadic trust is enabled; (iii) reduced local same-pole triangle purity under Filter ON even while active σ remains high. These arise from the interaction of SJT zones, exit, filtering, and (when on) trust updates.

Heterogeneity

3.18 We introduce substantial heterogeneity across several parameters:

- Ideological Camps: Bipolar initial opinions split the population into two opposed factions.
- Issue Salience: Diverse attention allocations drawn from a Dirichlet distribution mean agents prioritize issues differently.
- Cognitive Entrenchment: The entrenchment parameter β_i varies across agents, mixing dogmatic and open-minded individuals.
- Network Topology: The Barabási-Albert topology yields a heavy-tailed degree distribution: hubs are chosen as partners more often (higher exposure opportunity under random neighbor selection), not because they have stronger per-interaction influence.

Stochasticity

3.19 We integrate stochasticity to capture social noise and scheduling dynamics:

- The preferential attachment model generates network topologies stochastically.
- Shuffling shatters sequential order by randomizing agent activation at each step.
- Dialogue partner selection is stochastic, choosing randomly from active neighbors.
- The suppression filter operates probabilistically, intercepting rejection events with an efficacy rate of $E = 46\%$.
- Initial values for opinion, salience, and entrenchment are drawn randomly from their respective probability distributions.

Observation

3.20 We record the system's trajectory at the end of each step by collecting system-level and agent-level diagnostics:

- Global Polarization ($\sigma(t)$): The root-mean-square (RMS) of per-dimension opinion standard deviations across active agents (primary KPI; matches the exit story).
- Full-cohort Polarization ($\sigma_{\text{all}}(t)$): The same RMS construction over all agents; churned agents retain their last opinion vector (robustness check against survivorship).
- Active User Count (N_{active}): The sum of active nodes.
- Churn Rate: The cumulative fraction of the population that has permanently exited.
- Average Frustration: The mean frustration across active users.
- Platform Revenue ($R(t)$): Total advertising earnings under a stylized brand-safety step function of $\sigma(t)$.
- Trust Segregation Ratio: The ratio of mean ingroup trust to mean outgroup trust.
- Same-Pole Triangle Purity: Mean, over active agents, of each node's share of closed active–active–active triangles that are mono-pole (neighbor–neighbor edge required). This is local triangle purity by opinion pole, not a classical wedge-normalized clustering coefficient and not opinion-value weighted.

Details

3.21 This section details the mathematical formulations and structural rules governing initialization, dialogue, trust co-evolution, user exit, and platform monetization.

Initialization

3.22 At $t = 0$, the model builds an undirected, scale-free network G of $N_{\text{agents}} = 500$ nodes with attachment parameter $m_e = 3$.

3.23 To seed a pre-polarized society, we initialize opinion vectors $\vec{O}_i \in [-1, 1]^K$ ($K = 2$ issues) via a bipolar split. Each agent is assigned a camp pole $\pi_i \in \{-1, +1\}$ with equal probability. Independently for each issue dimension d, extremity is drawn from a uniform interval and scaled by the pole:

$$O_{i,d}(0) = \Pi_{[-1,1]}\Big(\pi_i \cdot U_{i,d}(0.4, 1.0)\Big) \tag{2}$$

where $U_{i,d}(0.4, 1.0)$ denotes an independent draw from $\mathcal{U}(0.4, 1.0)$ and $\Pi_{[-1,1]}$ clips opinions to $[-1, 1]$. The expected absolute extremity is 0.7, yielding two opposed, pre-polarized factions without additional Gaussian noise.

3.24 We draw each agent's salience vector $\vec{S}_i$ from a symmetric Dirichlet distribution:

$$\vec{S}_i \sim \text{Dirichlet}(\vec{\alpha}) \tag{3}$$

where $\vec{\alpha} = [1, 1]^T$ for $K = 2$ issues, ensuring that $S_{i,d} \in [0, 1]$ and $\sum_{d=1}^{K} S_{i,d} = 1.0$.

3.25 To model cognitive dogmatism, we draw agent entrenchment β_i from a normal distribution and floor it to prevent negative resistance:

$$\beta_i = \max\Big(\beta_{\min}, \nu_i\Big), \quad \nu_i \sim \mathcal{N}(\mu_\beta, \sigma_\beta^2) \tag{4}$$

with parameters set to default values: $\beta_{\min} = 0.5$, $\mu_\beta = 4.0$, and $\sigma_\beta = 1.5$.

3.26 All active network edges $(u, v) \in E$ begin with symmetric dyadic trust set to a uniform starting value:

$$T_{uv}(0) = T_0 = 0.5 \tag{5}$$

Every agent begins the simulation with zero behavioral frustration ($F_i(0) = 0$).

Input Data

3.27 The Dynamic-BABE model is a theoretical, self-contained simulation and does not ingest external empirical input data for its execution.

Submodels

Dialogue Dynamics and Opinion Updates:

3.28 Upon activation, agent i randomly selects an active neighbor j for dialogue, processing the incoming opinion through a multi-stage cognitive pipeline.

Step 1: Salience-Weighted Opinion Alignment (A_{ij})

3.29 We calculate the ideological distance between receiver i and sender j as a salience-weighted dot product:

$$A_{ij} = \sum_{d=1}^{K} O_{i,d} \cdot \left(\frac{S_{i,d} + S_{j,d}}{2} \right) \cdot O_{j,d} \tag{6}$$

This formulation captures a key psychological reality: mutually important issues—where salience is high—contribute disproportionately to perceived agreement or disagreement. Because opinions are bounded in $[-1, 1]$ and average saliences sum to 1.0, the alignment spans $A_{ij} \in [-1, 1]$, where $+1.0$ represents perfect alignment on highly valued topics and -1.0 marks absolute ideological clash.

Step 2: Perceived Influence Weight (w_{ij})

3.30 Adapting the BEBA model (Chen et al. 2021), we establish a baseline influence weight based on the receiver's cognitive entrenchment and opinion alignment:

$$w_{ij}^{\text{base}} = 1.0 + \beta_i A_{ij} \tag{7}$$

Under the Dyadic Trust system, relational trust modulates this weight, serving as a positive cognitive buffer:

$$w_{ij} = 1.0 + \beta_i A_{ij} + \gamma_T T_{ij} \tag{8}$$

where $\gamma_T = 0.3$ represents the trust influence coefficient and $T_{ij} \in [0, 1]$ represents the dyadic trust. This trust buffer mathematically models the relational buffer that keeps relationships intact during disagreement: high trust elevates the influence weight, preventing debate from deteriorating into hostile backfires. When trust co-evolution is disabled, the code omits the $\gamma_T T_{ij}$ term entirely (edges are not trust-initialized), yielding $w_{ij} = 1.0 + \beta_i A_{ij}$. Conceptually, shifting w_{ij} via T_{ij} is a source-credibility / relational-weighting mechanism layered on SJT zone classification, not a claim that trust itself is an SJT latitude.

Step 3: Social Judgment Zone Classification

3.31 Unlike the collective, synchronous averaging of Chen et al. (Chen et al. 2021), we translate this cognitive logic into a local, pairwise interaction paradigm. To track individual metrics like frustration, churn, and algorithmic interventions, we classify interactions using Social Judgment Theory (Sherif & Hovland 1961; Jager & Amblard 2005). The perceived influence weight w_{ij} dictates the interaction zone:

$$\text{Zone}(w_{ij}) = \begin{cases} 1 & \text{(Latitude of Acceptance)} & \text{if } w_{ij} > \tau_{\text{assim}} \\ 2 & \text{(Latitude of Non-Commitment)} & \text{if } 0.0 \leq w_{ij} \leq \tau_{\text{assim}} \\ 3 & \text{(Latitude of Rejection)} & \text{if } w_{ij} < 0.0 \end{cases} \tag{9}$$

where $\tau_{\text{assim}} = 0.3$ represents the assimilation threshold. We model these latitude boundaries as stylized, uniform thresholds across the population for mathematical tractability and to isolate co-evolutionary feedback. Empirical SJT latitudes are heterogeneous and dynamic; implications of the uniform assumption are discussed in Section .

Step 4: Algorithmic Moderation (Suppression Filter)

3.1 When active (Filter ON), the platform's Suppression Filter intercepts backfire (Zone 3) events with an efficacy rate of $E = 0.46$:[1]

$$w_{ij} \leftarrow 0.0, \quad \text{Zone}(w_{ij}) \leftarrow 2 \quad \text{with probability } E \tag{10}$$

Interception neutralizes the opinion repulsion, downgrading the conflict to Zone 2 (Ignore). Successful moderation also applies a frustration relief bonus $B_{hb} = 2$ (e.g., substituting a hostile item with lower-friction content).

Step 5: Opinion Update Rules

3.2 We update the opinion vector $\vec{O}_i$ depending on the final interaction zone:

- Zone 1 (Assimilation): The agent assimilates toward the sender using a DeGroot-style consensus update:

$$\vec{O}_i(t+1) = \Pi_{[-1,1]}\left(\vec{O}_i(t) + \mu_{ij}\left(\vec{O}_j(t) - \vec{O}_i(t)\right)\right) \tag{11}$$

  where the step size is entrenchment-damped and then clamped to a convex weight:

$$\mu_{ij} = \min\left(1.0, \, \max\left(0.0, \, \frac{w_{ij}}{1.0 + \beta_i}\right)\right) \tag{12}$$

  Without the clamp, high trust can push the raw ratio $w_{ij}/(1 + \beta_i)$ above 1; the clamp restores classical DeGroot convexity. Opinions are then clipped element-wise by $\Pi_{[-1,1]}$. A diagnostic (`scripts/mu_overshoot_diagnostic.py`; seeds 0–2) finds that the uncapped ratio exceeds 1 in 0% of assimilation updates under Trust OFF but $\approx 87.6\%$ under Trust ON; after the clamp, applied $\mu_{ij} > 1$ is 0% in both cases. Thus under Trust ON the assimilation step-size channel is often saturated at $\mu = 1$, so trust's main cognitive pathway is likely zone-shifting via $\gamma_T T_{ij}$ rather than larger DeGroot steps.

- Zone 2 (Non-Commitment): The interaction is ignored, leaving the opinion vector unchanged:

$$\vec{O}_i(t+1) = \vec{O}_i(t) \tag{13}$$

- Zone 3 (Backfire): The agent experiences ideological repulsion and pushes its opinion vector away from the sender:

$$\vec{O}_i(t+1) = \Pi_{[-1,1]}\left(\vec{O}_i(t) + \text{repulsion}_{ij}\left(\vec{O}_i(t) - \vec{O}_j(t)\right)\right) \tag{14}$$

  where

$$\text{repulsion}_{ij} = \min\left(1.0, \, \frac{|w_{ij}|}{1.0 + \beta_i}\right) \tag{15}$$

  subject to the same element-wise clipping operator $\Pi_{[-1,1]}$.

Behavioral Frustration and User Churn:

3.3 Unmitigated disagreements accumulate psychological costs, which we model as behavioral frustration F_i:

- Frustration Accrual (Backfire): Unmitigated backfires (Zone 3) increment frustration:

$$F_i(t+1) = F_i(t) + 1 \tag{16}$$

[1] The internal simulation flag remains `enable_bridge` for legacy continuity with the codebase; it toggles this Suppression Filter and does not implement constructive bridging ranking.

- Frustration Healing (Assimilation): Assimilative dialogue (Zone 1) relieves cognitive dissonance, healing frustration at a constant rate $H = 1$:

$$F_i(t+1) = \max\Big(0,\, F_i(t) - H\Big) \tag{17}$$

- Moderation Relief Bonus: Successful algorithmic interception spares the agent the backfire and grants a mental relief bonus $B_{hb} = 2$:

$$F_i(t+1) = \max\Big(0,\, F_i(t) - B_{hb}\Big) \tag{18}$$

3.4 At the end of each interaction, the agent exits if $F_i > T_c$ with $T_c = 15$ (i.e., exit at $F_i = 16$):

$$\text{Active}_i(t+1) = \begin{cases} \text{False} & \text{if } F_i(t+1) > T_c \\ \text{Active}_i(t) & \text{otherwise} \end{cases} \tag{19}$$

Churned users freeze instantly. They cease all dialogue, leave the active communication set, and generate zero advertising revenue.

Dyadic Trust Co-evolution:

3.5 Under the dynamic condition, dyadic trust T_{ij} co-evolves with dialogue outcomes:

$$T_{ij}(t+1) = \begin{cases} \min\Big(1.0,\, T_{ij}(t) + \delta_+\Big) & \text{if Zone}(w_{ij}) = 1 \text{ (Assimilation)} \\ \max\Big(0.0,\, T_{ij}(t) - \delta_-\Big) & \text{if Zone}(w_{ij}) = 3 \text{ (Backfire)} \\ T_{ij}(t) & \text{if Zone}(w_{ij}) = 2 \text{ (Ignore)} \end{cases} \tag{20}$$

where $\delta_+ = 0.02$ represents the slow accumulation of trust during agreement, and $\delta_- = 0.05$ captures the rapid collapse triggered by conflict. The resulting asymmetry ratio ($\delta_-/\delta_+ = 2.5$) mathematically formalizes Slovic's empirical trust asymmetry principle: social relationships are highly fragile, and conflict destroys trust much faster than cooperation builds it.

3.6 Because social ties naturally wither without active engagement, we apply a passive decay filter to every edge $(u, v) \in E$ at the end of each step:

$$T_{uv}(t+1) = T_{uv}(t) \cdot (1 - \lambda) \tag{21}$$

where $\lambda = 0.001$ represents the passive erosion rate.

Platform Revenue:

3.7 Step-wise revenue is a stylized scale on active users. In the main 2×2, bipolar initialization yields $\sigma(0) \approx 0.7 > \tau_{\text{cliff}} = 0.6$, so M_{safety} remains at the unsafe multiplier throughout and $R(t)$ tracks retention only:

$$R(t) = N_{\text{active}}(t) \cdot r_{\text{ad}} \cdot M_{\text{safety}}(t) \tag{22}$$

with $r_{\text{ad}} = \$1.00$ and

$$M_{\text{safety}}(t) = \begin{cases} M_{\text{safe}} = 1.0 & \text{if } \sigma(t) < \tau_{\text{cliff}} \\ M_{\text{unsafe}} = 0.4 & \text{if } \sigma(t) \geq \tau_{\text{cliff}} \end{cases} \tag{23}$$

Ablation M5 (multiplier fixed at 1.0) confirms the cliff mainly rescales dollars rather than driving the opinion paradox. Polarization $\sigma(t)$ uses population standard deviations (`np.std`, ddof $= 0$) as the RMS of per-dimension SDs over active agents:

$$\sigma(t) = \sqrt{\frac{1}{K}\sum_{d=1}^{K} \text{SD}\Big(\{O_{i,d}(t) : i \in \mathcal{A}(t)\}\Big)^2} \tag{24}$$

where $\mathcal{A}(t) = \{i \in V : \text{Active}_i(t) = \text{True}\}$. The model reporter may emit ∞ when outgroup trust is numerically zero; `stats_analysis.py` caps Trust Segregation at 10^6 as an analysis transform for tables and tests (not a change to the simulation). This polarization construction avoids the "data flattening" trap: averaging opinions across dimensions can cancel opposed extremes (e.g., $[+1, -1]$), whereas per-dimension dispersion preserved under RMS does not.

3.8 Ingroup and outgroup trust metrics classify an agent's pole by $\mathrm{sign}(\overline{O}_i)$, the sign of the mean opinion across issues. Agents with mean opinion exactly zero ($\mathrm{sign}(0) = 0$) are rare under bipolar initialization but, when present, are treated as a distinct zero pole for edge classification; we discuss this edge case as a limitation.

Table 1: Model Parameters and Default Values

Parameter Symbol	Description	Default Value	Source / Rationale
N_{agents}	Population size	500	Standard simulation population size
m_e	BA network attachment parameter	3	Scale-free social degree distribution
K	Number of opinion dimensions	2	Multidimensional issue space
β_{mean}	Mean of cognitive entrenchment	4.0	High entrenchment social media bias
β_{std}	SD of cognitive entrenchment	1.5	Cognitive heterogeneity in population
β_{min}	Minimum entrenchment floor	0.5	Prevent negative cognitive resistance
k_{step}	Dialogue partners selected per step	1	Micro-interaction conversational model
τ_{assim}	Assimilation threshold	0.3	Social Judgment Theory threshold
T_c	Churn frustration threshold	15	Theoretical exit tolerance ($F > T_c$)
H	Natural frustration healing rate	1	Healing per assimilation
B_{hb}	Suppression-filter healing bonus	2	Theoretical relief under Filter ON
r_{ad}	Base advertising rate per step	\$1.00	Retention scale unit
τ_{cliff}	Stylized revenue cliff	0.6	Inert in main 2×2 by design
M_{safe}	Safe revenue multiplier	1.0	Unused in main 2×2 trajectories
M_{unsafe}	Unsafe revenue multiplier	0.4	Constant scale under pre-polarized init
E	Suppression-filter efficacy	0.46	Exploratory interior knob (near 0.5; not fitted)
T_0	Initial dyadic trust weight	0.5	Relational neutrality starting state
γ_T	Trust buffer influence coefficient	0.3	Disagreement buffering coefficient
δ_+	Trust building step size (gain)	0.02	Slow relational capital accumulation
δ_-	Trust erosion step size (loss)	0.05	Slovic relational fragility parameter
λ	Passive trust decay rate	0.001	Under-maintenance erosion rate
t_{max}	Maximum simulation step limit	200	Conversational equilibrium epoch

3.9 Default values in Table 1 are exploratory theoretical knobs, not fitted to a proprietary log or tuned to a single churn target. $E = 0.46$ is an arbitrary interior value near one-half (not a calibrated rate); the OFAT grid uses $\{0.0, 0.2, 0.4, 0.6, 0.8, 1.0\}$ and therefore brackets the default without hitting it exactly. OFAT sweeps (`sensitivity.py`; `output/sensitivity_results.csv`; five replications; Trust OFF except when sweeping γ_T) show the qualitative filter–churn contrast is robust at interior points and fragile at boundaries: $E = 0$ restores Baseline-like exit; near-perfect $E = 1$ removes almost all repulsion and can drive Filter ON toward consensus ($\sigma \to 0$), so the paradox is an interior phenomenon. See Discussion.

Implementation and Replication

3.10 Dynamic-BABE is implemented in Python with Mesa 2.x (`mesa>=2.1,<3.0`; see `requirements.txt`), NumPy, and NetworkX. Replicate the main 2×2 with `batch_run.py`, then `stats_analysis.py`, `visualize.py`, and `python -m experiments.plot_experiments`. Optional checks: `scripts/mu_overshoot_diagnostic.py`, `scripts/check_reproducibility.py`. Public code: `https://github.com/TheCoolSam/BABEModel`. CSV condition labels retain the legacy name "Bridge Only" for the Filter Only cell; the paper maps that label to Suppression Filter ON.

Experimental Design

3.11 We map the processes above with a 2×2 design crossing the Suppression Filter and dyadic trust:

1. Baseline (Filter OFF, Trust OFF): Biased assimilation with no trust term in w_{ij} (code omits $\gamma_T T_{ij}$; edges are not initialized with trust).
2. Filter Only (Filter ON, Trust OFF): Suppression Filter on; same trust-off weight rule.

3. Trust Only (Filter OFF, Trust ON): Trust co-evolves from $T_0 = 0.5$; filter off.
4. Full Model (Filter ON, Trust ON): Both mechanisms active.

Model Hyperparameters and Topology

3.12 The model places $N = 500$ agents on a Barabási–Albert network (Barabási & Albert 1999) with $m = 3$. Opinions live in $K = 2$ issues with bipolar initialization: pole $\pi_i \in \{-1, +1\}$ with equal probability, and each issue set to $\pi_i \cdot U(0.4, 1.0)$ (clipped to $[-1, 1]$).

3.13 Entrenchment $\beta_i \sim \mathcal{N}(4.0, 1.5)$ with floor 0.5. When trust is on, influence shifts by $\gamma_T = 0.3$, gains $\delta_+ = 0.02$ on assimilation, loses $\delta_- = 0.05$ on backfire, and decays at $\lambda = 0.001$ per step. The Suppression Filter intercepts rejection-zone events with efficacy $E = 0.46$ and applies a two-point frustration relief. Defaults for E, T_c, and B_{hb} are theoretical knobs (not fitted); OFAT sweeps show the qualitative filter–churn contrast is robust at interior values and breaks mainly at identifiable boundaries ($E = 0$, near-perfect $E = 1$, very low β).

3.14 We run $R = 30$ replications per condition for $S = 200$ steps. Run i uses seed $42+i$. Initializing $\sigma(0) \approx 0.7$ above the stylized cliff $\tau_{\text{cliff}} = 0.6$ is intentional: the design evaluates demographic retention on an already pre-polarized platform, so M_{safety} stays at the unsafe scale in the main 2×2 and revenue tracks active users.

Stylized Facts and Qualitative Targets

3.15 Parameters are literature-motivated, not fitted to a proprietary platform log. We instead check that default Dynamic-BABE recovers directional patterns reported in related work:

Rejection-zone repulsion. Zone 3 contact moves opinions apart in the model (Sherif & Hovland 1961; Jager & Amblard 2005); field backfire evidence is contested (see Related Work).

Feed regime vs. deep attitudes. The filter changes exit/retention strongly while active-agent polarization stays high or rises, consistent in spirit with feed experiments where ranking shifts behavior more than core attitudes (Guess et al. 2023).

Fragile trust. Gain/loss asymmetry ($\delta_-/\delta_+ = 2.5$) follows Slovic (1993) and yields trust segregation when trust is enabled.

Exit under conflict. Frustration-driven churn implements Hirschmanian exit (Hirschman 1970) under unmoderated conflict.

3.16 These are qualitative anchors, not quantitative calibration targets.

Statistical Methodology

3.17 Final-step metrics across 120 runs use non-parametric tests because several KPIs are skewed or bounded. Because run i shares seed $42 + i$ across cells, primary pairwise inference is seed-paired Wilcoxon signed-rank on per-seed differences. Scientifically, the lead contrast is Baseline vs. Filter Only on retention (churn; equivalently active users) and active σ; step-wise revenue is the same retention contrast under fixed $M_{\text{unsafe}} = 0.4$, not an independent endpoint. Reported adjusted p-values come from a conservative Bonferroni secondary screen over the gated six-pair$\times$KPI matrix ($m = 62$ after excluding identically flat Trust-OFF trust KPIs); we do not claim a separate smaller primary family. Mann–Whitney U is retained only as a supplementary unpaired sensitivity. Cohen's d uses $(\bar{x}_{\text{Group1}} - \bar{x}_{\text{Group2}})/\text{SD}_{\text{pooled}}$ with Group order as labeled in each contrast (so d may be negative when Group2>Group1); for near-separated churn/revenue we emphasize mean differences and CIs, treating very large $|d|$ as a separation signal. We also report 95% bootstrap intervals (10,000 resamples). Trust Segregation is capped at 10^6 in analysis when outgroup trust is numerically zero (model reporters may emit ∞).

3.18 For Filter $\times$ Trust moderation we form seed-paired effects Δ_{alone} = (Filter Only) − (Baseline) and $\Delta_{\text{with trust}}$ = (Full Model) − (Trust Only), then apply Wilcoxon signed-rank to their difference, with a separate interaction Bonferroni family ($m = 7$ after excluding trust-KPI rows with $\Delta_{\text{alone}} \equiv 0$). Active Users, Churn, and Revenue are linearly dependent retention readouts in that family, not three independent moderation tests. Excluded trust-KPI rows are descriptive Full-vs-TrustOnly contrasts only.

Non-significant retention interactions are interpreted as no evidence of moderation at $n = 30$, not as equivalence; some raw p-values fall near 0.01–0.02 before correction and then fail Bonferroni. CSV outputs retain the legacy condition name "Bridge Only" for the Filter Only cell.

Results

4.1 Our 2×2 experiments cross the Suppression Filter (ON/OFF) with dyadic trust (ON/OFF). Each cell has 30 runs of 200 steps with seed $42 + i$ for run i. Primary pairwise inference uses seed-paired Wilcoxon signed-rank tests. The lead scientific contrast is Baseline vs. Filter Only on retention (churn) and active σ; revenue is a scaled retention readout under fixed M_{unsafe}. Adjusted p-values reported below come from a conservative Bonferroni secondary screen ($m = 62$ non-degenerate pairwise contrasts; flat Trust-OFF trust KPIs excluded). Mann–Whitney U is supplementary only. Cohen's $d = (\bar{x}_1 - \bar{x}_2)/\text{SD}_{\text{pooled}}$ follows the printed Group1 vs. Group2 order (negative d when Group2 is larger). For near-separated churn and revenue we emphasize mean differences and CIs; very large $|d|$ is a separation signal. SciPy's Wilcoxon W is the minimum of the two signed-rank sums ($W = 0$ indicates complete same-direction separation among nonzero paired differences). Primary polarization σ is over active agents; σ_{all} retains last opinions of churned agents. CSV condition labels retain the legacy name "Bridge Only" for Filter Only.

The Polarization Paradox: Retention vs. Active-User Dispersion

4.2 We begin with Baseline vs. Filter Only, isolating the Suppression Filter when trust is off. The tension between short-term retention and active-user polarization is the Polarization Paradox.

User Retention and Platform Revenue

4.3 In the unmoderated Baseline, cumulative churn reaches 39.18% (SD = 2.27%, 95% CI = [38.39%, 39.98%]) by step 200, leaving 304.1 active users on average (SD = 11.35, 95% CI = [300.10, 308.03]). Frustration among survivors is 0.9668 (SD = 0.1669, 95% CI = [0.9055, 1.0239]). Final step-wise revenue averages \$121.64 (SD = \$4.54, 95% CI = [\$120.04, \$123.21]) under the constant unsafe multiplier—a readability scale on retention, not an independent brand-safety mechanism in this design.

4.4 Activating the Suppression Filter (Filter Only) cuts churn to 0.38% (SD = 0.30%, 95% CI = [0.27%, 0.49%]; mean difference ≈ -38.8 percentage points; Figure 1). Seed-paired Wilcoxon $W = 0.0$, $p_{\text{adjusted}} < 1.1 \times 10^{-4}$ (near-complete separation). Active users rise to 498.1 (SD = 1.52, 95% CI = [497.57, 498.63]).

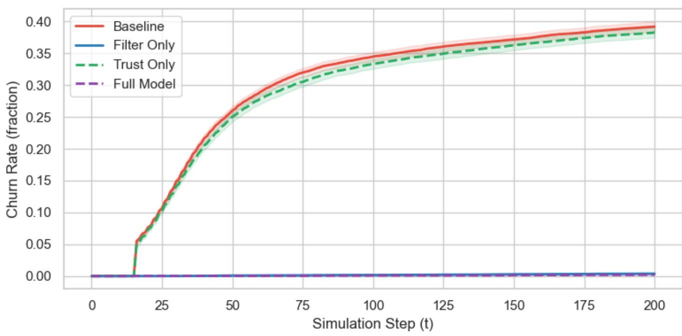

Figure 1: Cumulative user churn rate over time ($N = 500$, 30 runs per condition). Filter Only mean final churn is 0.38%; Full Model is 0.21%; Filter OFF conditions show high attrition ($\approx 39\%$). Error ribbons: 95% CIs.

4.5 Mean frustration among active users falls to 0.5819 (SD = 0.0576, 95% CI = [0.5617, 0.6024], $W = 2.0$, $p_{\text{adjusted}} < 3.5 \times 10^{-7}$, $d = 3.08$). Step-wise revenue rises to \$199.24 (SD = \$0.61, 95% CI = [\$199.03, \$199.45], $W = 0.0$, $p_{\text{adjusted}} < 1.1 \times 10^{-4}$). The $\approx$ \$78 mean increase is demographic retention under fixed $M_{\text{unsafe}} = 0.4$: because $\sigma(0) \approx 0.7 > \tau_{\text{cliff}} = 0.6$, the stylized cliff does not switch in the main 2×2.

Active-Agent Polarization and Full-Cohort Robustness

4.6 Active-agent σ rises from 0.9550 (SD = 0.0389, 95% CI = $[0.9409, 0.9683]$) in Baseline to 0.9904 (SD = 0.0042, 95% CI = $[0.9888, 0.9918]$) under Filter Only (mean difference +0.0353; $W = 12.0$, $p_{\text{adjusted}} < 8.1 \times 10^{-6}$, $|d| = 1.28$ with $d = -1.28$ under Group1=Baseline, Group2=Filter Only; Figure 2). Baseline exit removes extreme agents from the active set; the filter retains them.

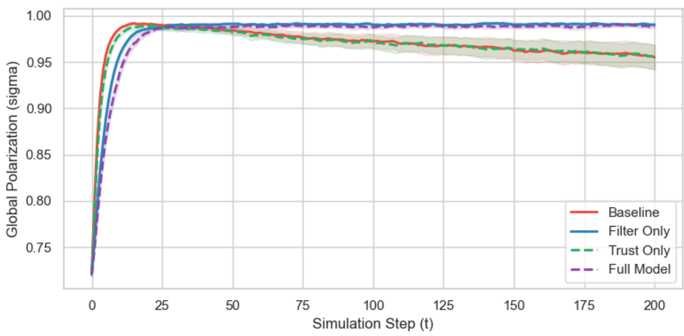

Figure 2: Active-agent polarization (σ) over time. Filter Only locks survivor polarization higher ($\sigma \approx 0.9904$) than Baseline ($\sigma \approx 0.9550$). This is an active-set / survivorship effect, not a full-population warming (see σ_{all}). Error ribbons: 95% CIs.

4.7 Full-cohort σ_{all} shows no Filter contrast: 0.9904 vs. 0.9905 ($W = 194.0$, $p_{\text{adjusted}} = 1.000$, $d = -0.004$). Retention effects remain; the claim that the filter raises polarization applies to the active-user KPI matching the exit story.

Trust Segregation

4.8 Under Trust Only, ingroup trust rises to 0.9808 (SD = 0.0056, 95% CI = $[0.9788, 0.9827]$) and outgroup trust falls to 0.0432 (SD = 0.0256, 95% CI = $[0.0348, 0.0528]$; Figure 3). Trust Segregation averages 31.28 (SD = 18.83, 95% CI = $[24.97, 38.21]$; median = 26.76, IQR = $[17.09, 40.66]$). Versus Baseline: $W = 0.0$, $p_{\text{adjusted}} < 1.2 \times 10^{-7}$ (complete same-direction separation; Baseline segregation is fixed at 1 when Trust OFF). These contrasts enable trust co-evolution; they are not Filter effects.

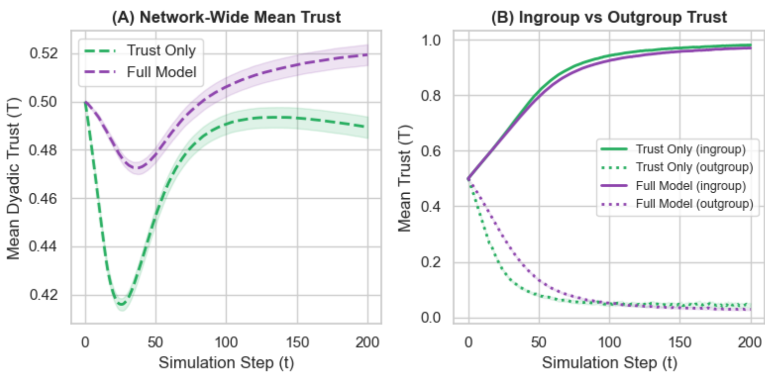

Figure 3: Dyadic trust dynamics. Panel A: mean trust under Trust Only and Full Model. Panel B: ingroup trust (near 0.98) vs. outgroup trust (near 0.04). Error ribbons: 95% CIs.

Moderation Analysis: Filter × Trust

4.9 Seed-paired contrasts:

$$\Delta_{\text{alone}} = \text{Metric(Filter Only)} - \text{Metric(Baseline)} \tag{25}$$

$$\Delta_{\text{with trust}} = \text{Metric(Full Model)} - \text{Metric(Trust Only)} \tag{26}$$

Wilcoxon tests on $\Delta_{\text{with trust}} - \Delta_{\text{alone}}$ use a separate interaction Bonferroni family of size $m = 7$ after excluding trust-KPI rows with $\Delta_{\text{alone}} \equiv 0$ (Table 2). Those excluded rows are descriptive Full-vs-TrustOnly contrasts on trust metrics, not Filter×Trust moderation. Same-pole triangle purity is included in the family (CSV column `Opinion_Clustering`); its interaction is non-significant ($W = 168$, adj. $p = 1$).

Table 2: Filter × Trust interaction (seed-paired). Interaction family $m = 7$ excludes trust-KPI rows with $\Delta_{\text{alone}} = 0$ (shown below the rule for transparency; adj. p n/a). Active Users / Churn / Revenue are dependent retention readouts.

Metric	Effect Alone (Δ_{alone})	Effect w/Trust ($\Delta_{\text{with trust}}$)	Interaction Cohen's d	W	Adj. p
Active Users	+194.0	+190.3	−0.3347	117.5	0.1250
Avg. Frustration	−0.3849	−0.3915	−0.0368	216.0	1.0000
Churn Rate	−38.80%	−38.05%	+0.3347	121.0	0.1523
Same-pole purity	−0.0486	−0.0446	+0.2086	168.0	1.0000
Polarization (σ)	+0.0353	+0.0336	−0.0469	208.0	1.0000
Polarization (σ_{all})	+0.0000	−0.0002	−0.0382	221.0	1.0000
Revenue	+\$77.60	+\$76.11	−0.3347	111.0	0.0871
Ingroup Trust†	0.0000	−0.0110	−3.2083	1.0	n/a
Mean Trust†	0.0000	+0.0300	+4.9482	0.0	n/a
Outgroup Trust†	0.0000	−0.0139	−0.9249	84.0	n/a
Trust Segregation†	0.0000	+5.3710	+0.4417	135.0	n/a

† Excluded from Bonferroni m: Trust OFF forces $\Delta_{\text{alone}} = 0$ (descriptive only).

4.10 Retention interactions (Active Users, Churn, Revenue—dependent readouts—and Frustration) remain non-significant after Bonferroni. Some raw p-values are ≈ 0.01–0.02 before correction and then fail the $m = 7$ family; with $n = 30$ we report no evidence of moderation, not statistical equivalence of Filter effects with and without trust. Active σ likewise shows no Filter×Trust interaction ($W = 208$, adj. $p = 1$).

Echo Chambers

4.11 Same-pole triangle purity (local closed active triples; Figure 4) averages 0.1316 in Baseline (SD = 0.0255, 95% CI = [0.1228, 0.1407]); Trust Only 0.1316 ($W = 226.0$, $p_{\text{adjusted}} = 1$). Filter Only lowers purity to 0.0830 (SD = 0.0152, 95% CI = [0.0776, 0.0884], $W = 0.0$, $p_{\text{adjusted}} < 1.2 \times 10^{-7}$, $|d| = 2.32$). Full Model: 0.0870 (SD = 0.0136, 95% CI = [0.0821, 0.0919], $W = 0.0$, $p_{\text{adjusted}} < 1.2 \times 10^{-7}$, $|d| = 2.19$). The filter can reduce local mono-pole triangles while active-agent σ stays high—another active-set / KPI nuance.

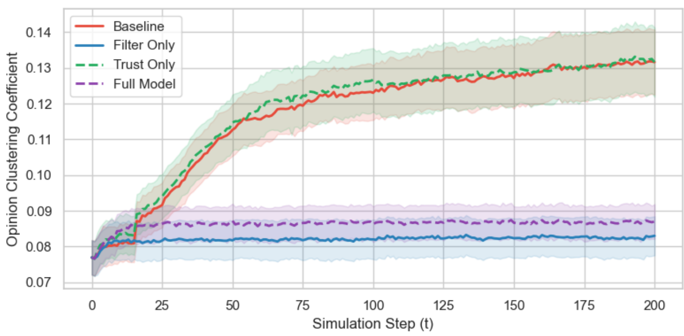

Figure 4: Same-pole triangle purity (local closed active triples). Filter ON conditions reduce local mono-pole triangles relative to Baseline while active-agent polarization remains high. Error ribbons: 95% CIs.

Mechanism Checks

4.12 Descriptive ablations (20 seeds; Figure 5; lower power than the main factorial). Relative to REF (churn 38.66%, $\sigma = 0.9579$), intercept+heal (M0) yields churn 0.39% and $\sigma = 0.9901$. Heal-only (M1) yields churn 5.82%. Drop without heal (M2) and intercept with $B_{hb} = 0$ (M3) are identical per seed under current code (both churn 20.53%, $\sigma = 0.9839$)—one mechanism identity, not two independent confirmations—and remain distinct from M0. Full Model without trust decay (M4) keeps near-zero churn. Setting the revenue multiplier to 1.0 (M5) leaves polarization/churn unchanged but scales revenue to the full active-user level.

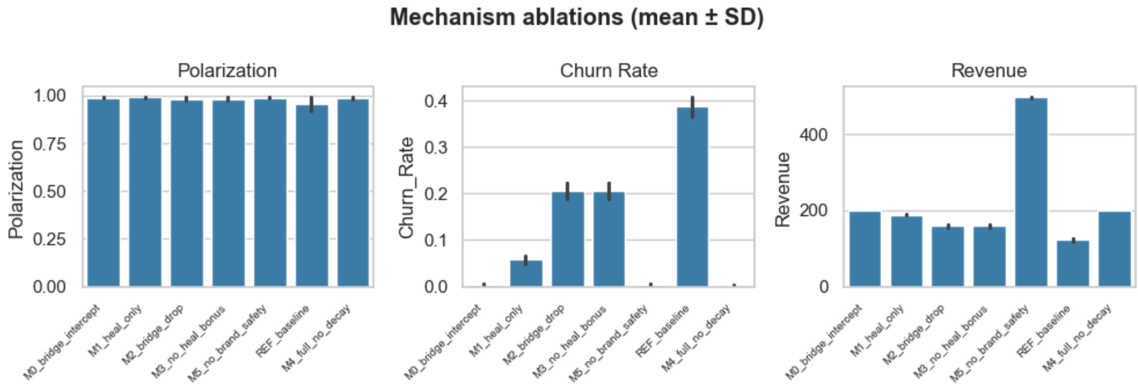

Figure 5: Mechanism ablations (20 seeds): mean Polarization, Churn Rate, and Revenue. Error bars: SDs. Descriptive only.

Baseline Model Comparison

4.13 On the same BA networks and seeds, HK bounded confidence yields lower final polarization (mean 0.6991) and no churn pathway. A lightweight biased-assimilation update without SJT zones or exit stays near-extreme (mean 0.9900) with zero churn. Dynamic-BABE Baseline jointly produces high polarization and substantial exit (Figure 6).

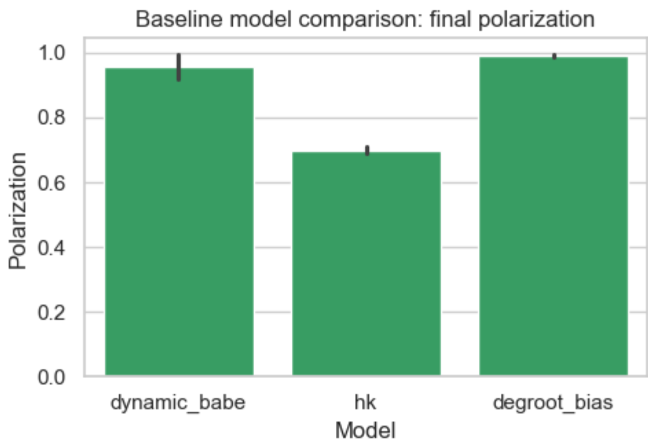

Figure 6: Final polarization under Dynamic-BABE Baseline, Hegselmann–Krause, and biased DeGroot-style updates (20 seeds).

Topology Robustness

4.14 Filter OFF/ON on Watts–Strogatz ($k = 6$, $p = 0.1$; 15 seeds; descriptive) preserves the qualitative direction: filter cuts churn while active-agent polarization rises, comparable to Barabási–Albert under the same seeds (Figure 7).

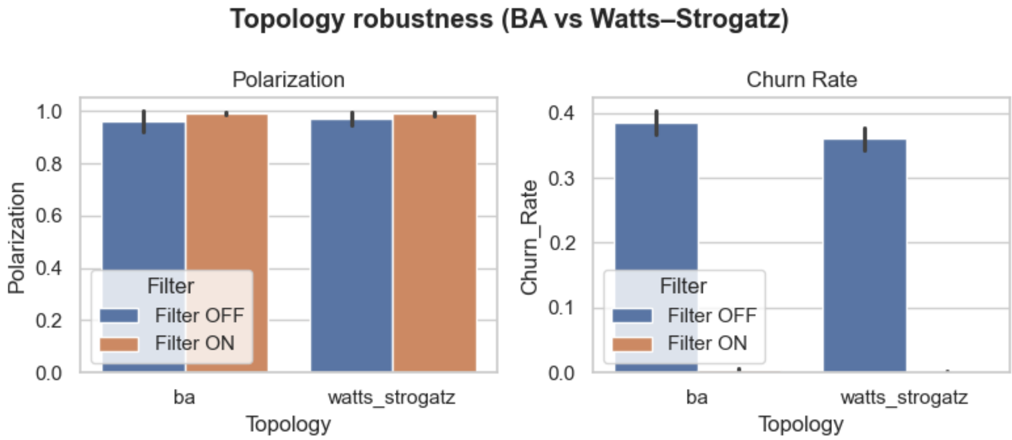

Figure 7: Filter OFF/ON contrasts on Barabási–Albert vs. Watts–Strogatz (15 seeds). Descriptive.

Discussion

5.1 We interpret the Polarization Paradox with Exit–Voice–Loyalty, discuss trust, state conditional implications, summarise robustness, and list limitations.

The Polarization Paradox: Exit, Voice, and Loyalty

5.2 Hirschman (1970) distinguishes Exit from Voice. In the Baseline, rejection-zone contact raises frustration and many agents exit—costly for the platform, but also removing some extreme agents from the active set.

5.3 The Suppression Filter intercepts many Zone 3 events and reduces exit. Retention and retention-scaled revenue improve; because $\sigma(0)$ already exceeds the stylized cliff, that revenue gain is demographic. The filter often replaces rejection with Zone 2 silence rather than constructive engagement, so it does not build Voice. Closing exit without opening repair leaves extreme agents active, sustaining high active-agent polarization while churn falls. Full-cohort σ_{all} does not rise, so the paradox is retention versus survivor dispersion / KPI misalignment—not population warming. The active-σ contrast is modest in absolute terms (+0.035 near an already high Baseline), so the substantive claim is survivorship preventing active-set moderation, not a large newly generated polarization wave.

Dyadic Trust

5.4 Trust is double-edged. Agreement raises ingroup trust (mean 0.9808 in Trust Only); conflict and asymmetry ($\delta_-/\delta_+ = 2.5$) following Slovic (1993) collapse outgroup trust (mean 0.0432). With passive decay, trust segregates (mean ratio 31.28; median 26.76). These patterns come from enabling trust co-evolution, not from the Filter contrast. Trust shifts w_{ij} before SJT zone classification (source-credibility weighting). Assimilation and repulsion step sizes are clamped to $[0, 1]$; under Trust ON the assimilation step often saturates at $\mu = 1$, so trust mainly acts as a zone buffer via $\gamma_T T_{ij}$. Under our defaults, Filter $\times$ Trust interactions on active σ and retention are non-significant after Bonferroni correction: we find no evidence that trust co-evolution attenuates the paradox (non-significance is not equivalence).

Uniform SJT Latitudes

5.5 Assimilation and rejection thresholds (τ_{assim}, the $w_{ij} < 0$ cut) are uniform across agents. That choice keeps the 2×2 interpretable but abstracts away person- and issue-specific latitudes that empirical SJT treats as heterogeneous and dynamic (Sherif & Hovland 1961). Heterogeneous thresholds could change how often the Suppression Filter fires and how exit concentrates among dogmatic agents; we leave that extension to future work and treat present zone boundaries as stylized knobs, not calibrated psychometrics.

Conditional Implications

5.6 Curation systems often optimise engagement and retention. Our results suggest those objectives need not track active-set dispersion or local same-pole structure among remaining users. If designers also care about cross-camp trust, suppression alone is an incomplete instrument. That statement is a conditional implication from a theoretical laboratory, not a prescription for any named platform.

Robustness

5.7 OFAT sweeps over β_{mean}, τ_{assim}, T_c, K, γ_T, E, and m_e (five replications per cell; $N = 500$; Trust OFF except when sweeping γ_T) preserve the qualitative filter–churn contrast at interior points. They do not substitute for the full 2×2. Boundaries matter: very low β; $E = 0$; and near-perfect $E = 1$, where Filter ON can collapse toward consensus once repulsion is almost entirely intercepted—so the paradox is an interior regime. See `sensitivity.py` and `output/sensitivity_results.csv`.

5.8 Additional checks (Results) are lower-power descriptive supplements: (i) placebo heal without intercept is weaker than intercept+heal; (ii) drop without heal and intercept with $B_{hb} = 0$ are the same stochastic process under current code (identical per seed) and distinct from M0; (iii) Dynamic-BABE differs from HK / biased-DeGroot baselines in linking suppression to exit; (iv) Filter OFF/ON remains directionally similar on Watts–Strogatz under our defaults.

Limitations and Future Work

1. Symmetric trust: $T_{ij} = T_{ji}$.
2. Static undirected topology: BA/WS graphs are fixed; real follower graphs are directed and rewiring.
3. Low-dimensional issues: $K = 2$.
4. Pole edge case: $\text{sign}(\overline{O}_i) = 0$ is a distinct pole (rare under bipolar init).
5. No field fit: Parameters are not estimated from a named platform dataset.
6. Stylized filter and revenue: Revenue $\approx$ retention under a pre-polarized cliff; not an audited production system.
7. Uniform latitudes: SJT thresholds do not vary by agent or issue.

5.9 Despite these limits, the model is a transparent testbed for how a suppression filter coupled to exit can misalign retention and active-user cohesion KPIs, and—separately—how fragile trust segregates on a fixed network.

Conclusion

6.1 Dynamic-BABE links opinion dynamics, dyadic trust, exit, and a stylized Suppression Filter. Under our defaults, the filter sharply improves retention (and retention-scaled revenue) while raising active-agent polarization by keeping extreme agents who would otherwise exit; full-cohort σ_{all} does not show that rise. Local same-pole triangle purity can fall even while active σ stays high—another KPI divergence under suppression. Trust co-evolution separately produces trust segregation. Seed-paired Filter $\times$ Trust tests show no evidence that trust attenuates the active-σ paradox or materially changes retention gains under these defaults; non-significance is not equivalence.

6.2 The practical implication is modest: platform KPIs need not move together. Retention, active-user dispersion, full-cohort dispersion, and same-pole triangle purity can diverge when disagreement is suppressed. Designs that only minimise visible conflict may miss relational outcomes that matter for long-run tolerance. Under our defaults we find no evidence that enabling trust co-evolution resolves the active-set trade-off; whether other contact or repair regimes could do so remains an open modelling question beyond the present 2×2.

Code and Data Availability

6.3 The Dynamic-BABE implementation, batch scripts, analysis pipeline, and figure generators are publicly available at `https://github.com/TheCoolSam/BABEModel`. We intend to deposit a reviewed release on the CoMSES Net Computational Model Library so that a permanent handle can accompany the published article.

Author Biographies

Samuel Botshtein. Samuel Botshtein is an independent researcher and the lead software engineer for Dynamic-BABE. He is interested in complex systems as they relate to applied physics, especially how local interaction rules produce large-scale patterns that can be studied with simulation. On this project he designed and implemented the agent-based model, experiment pipeline, and analysis tooling in Python, and helped prepare the manuscript for the Journal of Artificial Societies and Social Simulation. He continues to develop work at the intersection of computational modeling, networked systems, and open scientific software.

Aurnob Ghose. TODO: author bio (approximately 100 words). Independent researcher.

Felix Pipal. Felix Pipal is an independent researcher interested in complex systems and how online platforms shape group behavior. He is especially drawn to questions about polarization, trust, and how design choices in feeds and moderation change what people see and whether they stay. He contributed to the Dynamic-BABE project through model discussion, writing, and preparing the manuscript for the Journal of Artificial Societies and Social Simulation. He is building experience with agent-based modeling and computational social science, and plans to keep studying socio-technical systems through simulation and related methods.

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Notes

[1]CoMSES upload to be completed by the authors prior to publication.

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